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Rehabilitation of Swallowing Disorders

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SUMMARY PEARLS

- Swallowing is a rapid, yet complex sequence of both voluntary and reflexive movements comprised of multiple phases.
- Cortical and subcortical regions of the brain play an integral part in mediating swallow function, especially the voluntary phases of the swallow (oral preparatory and oral propulsive).
- Dysphagia can occur because of a wide variety of functional or structural deficits and sensory or motor impairment and can lead to serious complications, including dehydration, malnutrition, pneumonia, or airway obstruction.
- A patient-centered, multidisciplinary approach to assessment and therapeutic intervention for dysphagia often includes an instrumental assessment of swallowing to correctly identify the specific biomechanical components of the individual's swallow function to target for compensatory or restorative interventions.
- The goal of treatment for dysphagia is to restore swallowing skills and maximize quality of life while continuously maintaining adequate nutrition, hydration, means of medication administration, and safe (aspiration-free) swallowing for the individual.

The normal swallow is a complex and dynamic physiologic process involving coordination between multiple sensorimotor systems to move food or liquid bolus through the mouth, throat, and esophagus into the stomach while protecting the airway. Neurologic control of swallowing involves an intricate neuronal network with both voluntary and reflexive components, which regulate more than 30 paired muscles with sensory and motor information traveling over six cranial nerves.⁹⁷ Swallowing impairment, known as dysphagia, can result from a wide variety of functional or structural deficits of the oral cavity, pharynx, larynx, or esophagus and can occur at any time across the lifespan. This chapter provides a broad overview of swallowing, including the physiology and pathophysiology of eating and swallowing, as well as assessment and treatment methods.

Swallowing Physiology

Swallowing comprises reflexive and voluntary movement patterns that change as individuals develop, but the overall sequencing of biomechanical events remains stable over the lifespan. The act of swallowing has been described using the four-stage sequential model for drinking and the process model for eating. The four-stage model (Fig. 3.1A), which includes the oral preparatory, oral propulsive, pharyngeal, and esophageal stages, adequately describes the biomechanics and bolus movement of liquids; however, this model cannot represent the bolus movement and process of eating food. To describe the mechanism of eating and swallowing solid food, the process model of feeding was developed (see Fig. 3.1B).^{33,86}

Oral Phase—Drinking

Preparatory Stage

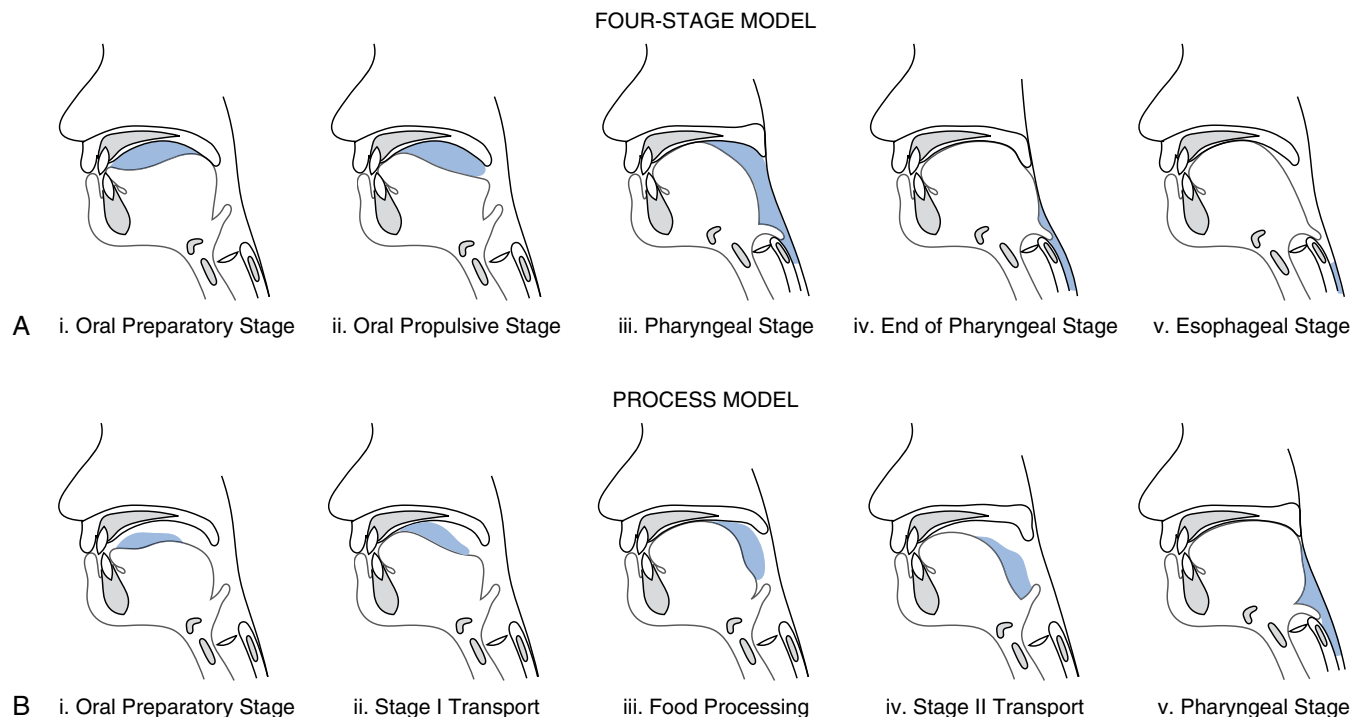
The oral preparatory stage of swallowing is a voluntary phase that occurs within the oral cavity. The bolus is taken into the mouth and held in the anterior portion of the floor of the mouth on the tongue surface against the hard palate. Simultaneously, the lips maintain a seal to prevent anterior spillage. The oral cavity seals posteriorly by contact of the soft palate and tongue to prevent the bolus from leaking into the oropharynx before swallowing initiation.⁷⁰

Propulsive Stage

The goal of the oral propulsive stage is to successfully transport the bolus from the oral cavity to the pharynx. During this voluntary phase of swallowing, the tongue tip elevates and contacts the alveolar ridge of the hard palate just behind the upper incisors. In contrast, the posterior portion of the tongue depresses to open the back of the oral cavity. As the tongue surface elevates, the area of tongue-palate contact expands from anterior to posterior, propelling the bolus into the pharynx. The pharyngeal phase of the swallow normally begins during oral propulsion when drinking.⁷⁰

Oral Phase—Eating (Process Model of Feeding)

The four-stage model has limited utility for describing the process of normal eating in humans because of the overlap among the oral preparatory, oral propulsive, and pharyngeal stages.^{25,33,86} Therefore the process model of feeding, originating in studies of mammalian



• **Fig. 3.1** The four-stage model for swallowing and the process model for swallowing. (A) The four-stage model for swallowing. (i) The bolus is held on the tongue's surface against the hard palate, and the oral cavity is sealed. (ii) Tongue movement serves to transport the bolus from the oral cavity to the pharynx. (iii) The swallowing reflex is triggered, the hyoid and larynx are elevated, the pharynx shortens, and pharyngeal constrictors contract sequentially. (iv) The pressure of the descending bolus distends the upper esophageal sphincter and allows entry of the bolus into the esophagus. (v) After the bolus travels into the esophagus, esophageal peristalsis is triggered reflexively. (B) The process model for swallowing. (i) Solid food is positioned on the tongue. (ii) Solid food is transferred to the postcanine region. (iii) Food is masticated and softened by salivation. (iv) The bolus is propelled into the oropharynx and may briefly accumulate. (v) The swallowing reflex is initiated, and the bolus is propelled through the pharynx to the esophagus.

feeding,¹⁰⁸ was adapted for feeding in humans.⁸⁶ When healthy humans eat solid food, triturated (chewed and moistened) food commonly passes through the faucial arches for bolus formation in the oropharynx and vallecula several seconds before pharyngeal stage swallow initiation. Additional portions of food may also pass into the oropharynx and accumulate there while food remains in the oral cavity, and chewing continues.⁷⁰

Stage I Transport

Once food has been taken into the mouth, the tongue carries the solid food to the postcanine region and then rotates laterally, placing food on the occlusal surface of the lower teeth for food processing.⁷⁰

Food Processing

Immediately following stage I transport, food processing occurs. During this process, food is masticated and softened by salivation until the consistency is optimal for swallowing. Throughout this rhythmic process, the tongue moves mediolaterally and rotates on its long axis to allow food to remain on the occlusal surfaces of the teeth. The cyclic movement of the jaw varies among food consistencies and is tightly coordinated with the movements of the tongue, cheek, soft palate, and hyoid bone, which permits open communication between the oral cavity and pharynx. Because there is no sealing of the posterior oral cavity during eating, the

airway remains open for breathing, and the jaw and tongue movements pump air into the nasal cavity through the pharynx, delivering aroma to chemoreceptors in the nose.^{5,36,85}

Stage II Transport

Once processed, food is placed on the tongue's surface and propelled back through the faucial pillars to the oropharynx. The tongue primarily drives this action without the need for gravity. Stage II transport of solids is similar to the oral propulsive stage for liquids. However, prepared foods often accumulate on the tongue's pharyngeal surface and vallecula while chewing continues in the oral cavity. The aggregated bolus in the oropharynx continues to enlarge by subsequent stage II transport cycles, lasting from a fraction of a second to 10 seconds in normal individuals before the initiation of the pharyngeal stage of the swallow.⁷⁰

Pharyngeal Stage

As the bolus is propelled past the anterior faucial pillars, mechanoreceptors of the tongue, epiglottis, and pyriform fossa are activated, which trigger the pharyngeal phase of the swallow.⁴⁰ The pharyngeal stage of the swallow is a rapid sequence of events that occurs within 1 second. This stage of the swallow has two essential goals: (1) passage of food through the pharynx and upper esophageal sphincter (UES) to the esophagus and

(2) preventing airway compromise (i.e., protecting the larynx and trachea during food transport to prevent the food from entering the airway).⁷⁰

During the pharyngeal stage, the soft palate elevates and retracts as the pharyngeal wall compresses along the soft palate, closing the nasopharynx and preventing bolus regurgitation into the nasal cavity. As the bolus passes into the pharynx, the base of the tongue retracts posteriorly, pushing the bolus against the pharyngeal wall. The pharynx shortens vertically to reduce the volume of the pharyngeal cavity while the pharyngeal constrictors contract sequentially from top to bottom, propelling the bolus downward toward the esophagus.⁷⁰

As the bolus passes through the pharynx, the true vocal folds adduct to seal the glottis (the space between the vocal folds), the false vocal folds approximate, and the arytenoids tilt forward while the epiglottis inverts for arytenoid-epiglottic contact before opening of the UES.^{81,94} The hyolaryngeal complex is pulled upward and forward by contraction of the suprahyoid and thyrohyoid muscles to tuck the larynx under the base of the tongue. At the same time, the epiglottis inverts to seal the laryngeal vestibule.⁶⁰ Together, these mechanisms prevent aspiration of foreign materials into the trachea before and during swallowing.

The UES opening is necessary for entry of the bolus into the esophagus. The UES is closed at rest because of tonic muscle contraction^{14,28}; however, it opens because of three crucial factors: (1) relaxation of the cricopharyngeus muscle; (2) contraction of the suprahyoid and thyrohyoid muscles, which pull the hyolaryngeal complex forward, opening the sphincter; and (3) pressure of the descending bolus, which serves to distend the UES.⁹⁶ These mechanisms are essential for bolus entry into the esophagus.

Esophageal Stage

Once the bolus has passed through the UES, it is propelled down the esophagus by peristalsis. The two main parts of the peristaltic wave are an initial wave of relaxation that accommodates the bolus and a wave of contraction that propels it. In the upright position, gravity assists peristalsis. The lower esophageal sphincter (LES), held closed by tonic muscle contraction between swallows, then relaxes during the swallow because of pressure of the descending bolus, allowing passage of the bolus into the stomach.⁷⁰

Coordination of Swallowing and Breathing

Swallowing and breathing share a common anatomic pathway; thus the processes of eating, swallowing, and breathing must be tightly coordinated to protect the airway. Muscles of the lips, face, tongue, pharynx, larynx, and esophagus are active during breathing and swallowing. Therefore they serve multiple purposes, including maintenance of airway patency, airway protection, and bolus propulsion. Supporting this idea is the evidence of a tight neural link between swallowing and breathing, which has resulted from the work of many neurophysiologists who have identified and detailed specialized neural networks in the brainstem and cortex.^{21,41,73,75} Further, single neurons within these medullary networks have been found to demonstrate multifunctionality in the control of both behaviors.⁴¹

In healthy adults, swallowing supersedes respiration,^{72,79,80} typically occurring during the expiratory phase of respiration

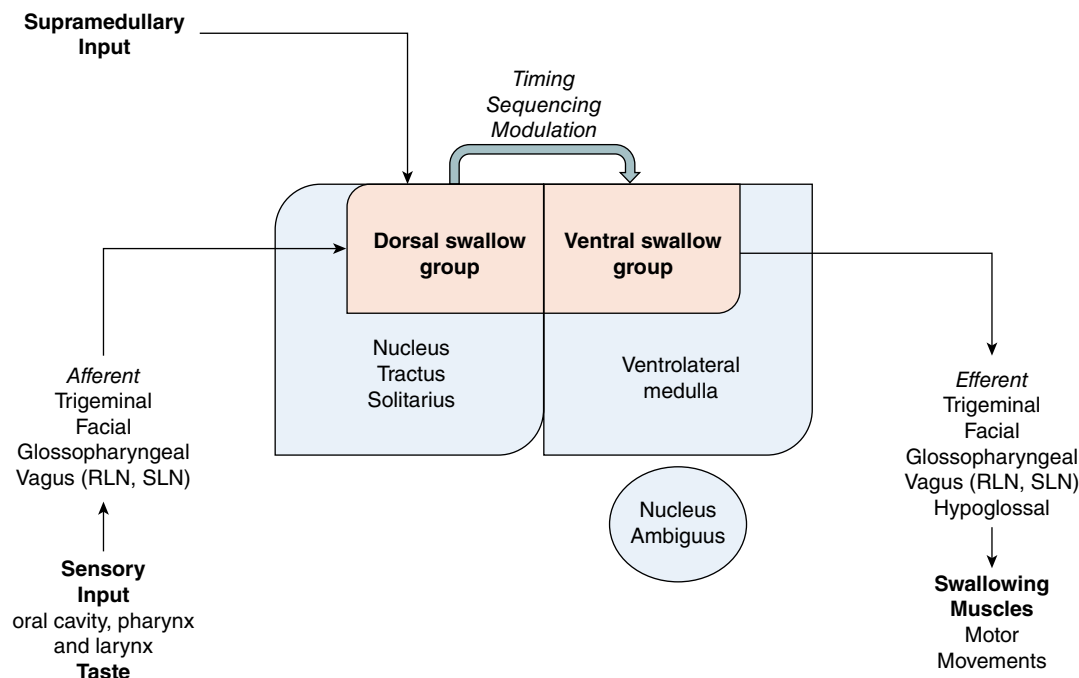
between middle and lower lung volumes⁷² with an obligate pause in respiration to accommodate swallowing.⁸⁰ Initiation of the swallow under these conditions poses a distinct physiologic advantage for protective mechanisms, such as hyolaryngeal anterior-superior movement, airway closure, and pharyngoesophageal segment opening.⁹ Onset of the respiratory pause, which averages from 1 to 1.5 seconds during liquid swallows, has also been associated with airway protection through a momentary medialization of the true vocal folds followed by complete vocal fold adduction during anterior-superior hyolaryngeal motion.⁶⁵ Typically, respiration resumes during the descent of the larynx at the later stages of the swallow and is marked by a brief exhalation.⁶⁵

Disruption of the tight neural coupling between the central control of breathing and swallowing has been observed in conditions such as aging, neurologic or neuromuscular disease, and cancers of the head and neck, resulting in variation in the stability of respiratory-swallowing coordination.^{6,24,35,67} The coupling of respiratory-swallowing phase patterning also changes based on the characteristics of the swallowing task. Cued vs. spontaneous cup drinking, postural maneuvers, variation in bolus volume, as well as modifications of food and liquid texture^{4,23,69,79} have been shown to alter the phase (i.e., inspiratory, expiratory) in which the swallow occurs as well as the timing and duration of the respiratory pause during the swallow.⁶⁶ These alterations in respiratory-swallowing phase patterns have been associated with increases in the presence of aspiration and swallowing impairment, so clinicians should not only be aware but also identify and optimize peripheral inputs to stabilize aberrant respiratory-swallowing patterns.

Neurophysiology of Swallowing

Swallowing is a dynamic, multilayered sensorimotor process involving a complex neuronal network. The precise pattern of sequential muscle activation and inhibition is dependent on portions of the supratentorial (cortical and subcortical), infratentorial (brainstem), and peripheral nervous system (motor and sensory).⁹⁷ The central nervous system mediates the swallowing process throughout each phase. During the swallow's oral preparatory and oral transport phases, cortical regions such as the primary and secondary sensorimotor cortices are especially active. These phases consist mainly of voluntary behaviors (i.e., bolus preparation and mastication). The pharyngeal phase involves several cortical and subcortical structures. These include the primary and secondary sensorimotor cortices, insula, anterior and posterior cingulate cortices, basal ganglia, amygdala, hypothalamus, and substantia nigra.^{27,100} Inhibition of the swallow is important at times, such as the oral preparatory and oral transport phases of the swallow, as it would be inefficient and ineffective to trigger a pharyngeal swallow during bolus preparation.⁹⁷ Cortical representation of swallowing in healthy individuals is bilateral but asymmetric, especially in the motor and premotor cortices,¹⁰⁰ thought to be caused by a dominant swallowing hemisphere, which has important rehabilitation implications particularly in individuals with conditions such as unilateral stroke.

The second level of neurophysiologic control for swallowing involves the medulla within the brainstem. This region is active during swallowing, most notably during the pharyngeal and esophageal phases. The medullary central pattern generator (CPG) regulates and coordinates these phases of the swallow



• **Fig. 3.2** Neuroanatomy of swallowing. The medullary central pattern generator (red) regulates and coordinates the phases of the swallow. RLN, Recurrent laryngeal nerve; SLN, superior laryngeal nerve. (From Langton-Frost N, Murphy K, Bahouth MN, et al. A scoping review of post-stroke dysphagia: evaluation and interventions. *Curr Phys Med Rehabil Rep.* 2024;12:241-249.)

(Fig. 3.2). The CPG is bilaterally represented; evidence of this is seen in studies in which unilateral stimulation triggers a swallow on the ipsilateral oropharyngeal side.^{21,22} Within this CPG, two groups of interneurons are integrated: the dorsal swallowing group (DSG) and the ventral swallowing group (VSG).⁴¹ When sensory signals from the oropharyngeal area (via cranial nerves [CN] IX and X) reach the nucleus of the solitary tract and reticular formation, they excite interneurons housed within the DSG, initiating motor signaling to the neighboring VSG, which then sequences and coordinates the motor input. The signal from the VSG is then passed on to the hypoglossal motor nucleus (CN XII) and nucleus ambiguus (CN X), which initiates motor activity in the swallowing musculature.⁴¹ These motor and sensory neurons responsible for swallowing are located in the same medullary sites as neurons that control respiration, enabling efficient coordination of swallowing and breathing.

Muscle movements for swallowing are controlled by neurons localized within the brainstem. These include the trigeminal (CN V), facial (CN VII), and hypoglossal (CN XII) motor nuclei, the nucleus ambiguus (NA) (CN IX and X) and the dorsal motor nucleus of the vagus nerve (CN X), as well as cervical spinal neurons (C1 and C3).^{22,26,74} However, not all these motoneurons participate equally during swallowing, with the primary motor nuclei involved in swallowing being the hypoglossal motor nucleus (CN XII) and the NA. These nuclei provide innervation for the intrinsic and extrinsic muscles of the tongue as well as the pharynx, larynx, and upper esophagus, making them crucial for swallowing.

Neurons responsible for sensory regulation are also located within the brainstem. General sensory information from the oral, pharyngeal, and esophageal mucosa is sent to the nucleus of the

solitary tract and the neighboring reticular formation. The tongue is the source of a tremendous amount of sensory input; therefore multiple cranial nerves are responsible for the relay of information such as temperature, texture, size, and volume of the bolus. General sensory information for the anterior two-thirds of the tongue is provided by the trigeminal nerve (CN V), while the posterior one-third is innervated by the glossopharyngeal nerve (CN IX). Special sensory information, such as taste, has been shown to alter swallowing physiology in oral and pharyngeal phases^{57,63} with innervation provided by CN VII (facial, anterior two-thirds of the tongue) and CN IX (posterior one-third of the tongue).⁸

Because swallowing is a complex, tightly coordinated process that involves movements within the oral cavity, pharynx, larynx, and esophagus, any impairment within this system (structural or neural) can result in swallowing impairment.

Dysphagia

Dysphagia is simply defined as difficulty swallowing. It results from various diseases and disorders^{7,48} and can lead to significant complications such as dehydration, malnutrition, pneumonia, or airway obstruction. It is also associated with diminished quality of life⁹⁸ and mortality.⁷¹ Functional or structural deficits of the oral cavity, pharynx, larynx, and esophagus can lead to dysphagia, and swallowing disorders are frequently described in terms of stage affected: oral, pharyngeal, or esophageal. Regardless of the site, it is helpful to consider whether a given swallowing impairment affects food transport (preparation and propulsion of the bolus), airway protection (prevention of laryngeal penetration/aspiration), or both because of the implications for rehabilitative intervention.

Oral Phase

The oral preparatory and oral transit stages of the swallow involve movements of the lips, tongue, cheeks, and jaw for the purpose of mastication, bolus preparation, and bolus transport. Weakness or uncoordinated movements of any of these structures can result in impairment. For example, a weak lip seal may result in anterior spillage of the bolus, especially with liquids. Impairments of the tongue and cheeks may lead to reduced bolus formation or bolus retention in the lateral sulcus.⁷⁰ Premature spillage of the bolus from the oral cavity into the throat may arise from impaired tongue base function. Additionally, impaired lip control for sucking or insufficient velopharyngeal seal, as well as reduced mastication, can result from structural abnormalities such as cleft lip/palate or malalignment of the teeth, respectively.

Pharyngeal Phase

Pharyngeal impairments are difficult to visualize during clinical assessment, necessitating instrumental evaluation. Although signs and symptoms of pharyngeal disorders can be observed at the bedside, such as coughing, choking, and multiple swallows, it is impossible to determine the cause of these behaviors without instrumental assessment. Abnormalities in the pharyngeal phase may include impaired swallow initiation, ineffective bolus propulsion, pharyngeal retention, and airway invasion of the bolus (laryngeal penetration and/or aspiration). Although retention of all or part of the bolus in the pharyngeal recesses can occur because of impaired tongue base retraction, similar findings can be produced by weakness of the pharyngeal constrictor musculature. Reduced epiglottic inversion is often seen on instrumental evaluation, which is a passive movement dependent upon bolus pressures, base of tongue retraction, and hyolaryngeal elevation.⁷⁰ When the epiglottis does not invert during the swallow, it may act as a physical barrier, contributing to retention of the bolus in the valleculae after swallowing. Cervical osteophytes are a structural abnormality common in older adults, which may contribute to dysphagia by narrowing the food pathway and directing the bolus toward the airway.¹⁹ Other structural impairments to consider are webs and strictures in the pharynx or sphincters.

Airway protective mechanisms are crucial for swallowing safety; however, they are not always effective. For dysphagia rehabilitation, *laryngeal penetration* is defined as passage of material into the larynx but not through the vocal folds, while *aspiration* is defined as passage of material through the vocal folds.⁷⁰ Failure of protective mechanisms—as demonstrated by reduced laryngeal elevation, incomplete laryngeal vestibule closure, or inadequate vocal fold closure caused by weakness, paralysis, or anatomic fixation—can result in laryngeal penetration or aspiration.⁷⁰ Laryngeal penetration and/or microscopic aspiration can be observed in healthy individuals. Still, aspiration visible on instrumental examination is pathologic and often associated with an increased risk of aspiration pneumonia or airway obstruction.⁶⁴ Typically, a robust reflexive cough or throat clear is the response to aspiration, but laryngeal sensation is often impaired in those with dysphagia.³⁰ Silent aspiration, or aspiration in the absence of observable response, has been reported in 25% to 30% of patients referred for dysphagia evaluation.^{30,54} The consequences of aspiration are highly variable and tolerated differently among individuals based on factors such as quantity of the aspirate, depth of aspirated material in the airway, physical properties of the aspirate, the

individual's pulmonary clearance efficiency, general pulmonary status, and overall health condition.⁸⁴

Impaired UES opening is another finding commonly observed on instrumental exam, which can cause partial or even total obstruction of the foodway, increasing the risk of aspiration after the swallow.⁷⁰ Failure of the UES to open can be structural, caused by a web or stricture, or functional, because of muscle weakness. It is sometimes difficult to differentiate these conditions, so empiric dilation is recommended.

Esophageal Dysphagia

Esophageal dysphagia often results from either a physical obstruction or motility disorder that prevents food and liquids from easily passing through the esophagus. Esophageal motor disorders include conditions of hyperactivity, hypoactivity, or incoordination of the esophageal musculature and can often be asymptomatic.⁷⁶ These conditions may contribute to ineffective peristaltic movement and retention of swallowed material in the esophagus with subsequent regurgitation into the pharynx, leading to an increased risk of aspiration. Some causes of esophageal dysphagia include achalasia, esophageal spasm, stricture, esophageal tumors, foreign bodies, scleroderma, and gastroesophageal reflux disease. Diagnosis of esophageal dysphagia requires instrumental assessment, and individuals should be referred to a gastroenterologist for further evaluation.

Conditions Resulting in Dysphagia

Dysphagia can occur because of multiple disorders. For that reason, careful determination of the underlying etiology of dysphagia may necessitate a broad diagnostic approach, including careful history, neurologic examination, and relevant studies (e.g., radiologic, electromyography, lab, biopsy). One of the most common causes of dysphagia is stroke. Dysphagia has been reported in 65% of stroke patients, only half of whom are symptomatic.¹³ Many recover within the first 2 weeks, but dysphagia can be severe and persistent. Brainstem lesions often result in the most severe forms of dysphagia, given the proximity to major swallow centers.^{41,103} Dysphagia as a result of a stroke may impact all phases of the swallow with signs and symptoms such as loss of oral control, reduced laryngeal elevation, insufficient UES opening, vocal fold weakness, dyscoordination of the swallow, impaired bolus propulsion, and diminished airway protection. More severe signs and symptoms of dysphagia are observed following bilateral cerebral lesions, as there is bilateral cortical representation for swallow function. By contrast, brainstem motor nuclei innervate only ipsilateral muscles, so lesions of cranial nerves or their nuclei can result in unilateral sensory or lower motoneuron dysfunction.³¹ Regardless of the type of stroke, pneumonia causes the highest attributable mortality of all medical complications following a stroke and may affect up to as many as one-third of all stroke survivors.² Most available data suggest that poststroke pneumonia is often caused by aspiration.²

Dysphagia is very often one of the first noted symptoms of neurologic disease.⁵⁸ Oral dysphagia is common in Parkinson disease and is characterized by tremor, dyskinesia, and bradykinesia in the lips, tongue, jaw, and larynx, which impairs oral and pharyngeal swallow function.¹ Symptoms of dysphagia in advanced disease include drooling, jaw rigidity, impulsive feeding behavior, and dysfunction in the UES and LES. In Alzheimer disease, agnosia for food within the oral cavity and subsequent

oral holding and incoordination of swallowing are common. Dysphagia in motoneuron diseases such as amyotrophic lateral sclerosis occurs because of degeneration of motoneurons in the brain and spinal cord, resulting in weakness in muscles of mastication, respiration, and swallowing. Early signs of bulbar involvement may include lingual and labial weakness, with bulbar involvement associated with a worse prognosis secondary to the high risk of pulmonary aspiration and malnutrition. Progressive loss of muscle function can result in severe dysphagia, so close monitoring of the individual's swallowing function is recommended. Bulbar and facial muscles are also affected in approximately one-third of those with myasthenia gravis, leading to weakness of mastication, nasal regurgitation, and other symptoms such as poor gag reflex, limited palate elevation, and tongue weakness.

Swallowing impairment is common in the case of tumors of the oral cavity, pharynx, and larynx because of tissue loss or dysfunction. Treatment for these tumors often involves surgical excision of structures, chemotherapy, or radiation therapy, all of which negatively impact the swallow. The tumors themselves can decrease mobility or cause fixation of the tongue, soft palate, pharynx, or larynx while also affecting the afferent fibers from the mucosa. Together, these changes can lead to decreased transport of food and liquid boluses and decreased airway protection.

As mentioned, therapies such as radiation therapy can contribute to dysphagia. Radiation can lead to fibrosis of the oral cavity, pharynx, and larynx, with effects sometimes evident immediately but also much later in the course of treatment. In the acute stages, there may be changes in the mucosa (e.g., erythema, mucositis, ulceration), taste buds can have decreased or altered acuity, and there is reduced production or thickened saliva. Late effects include fibrosis resulting in decreased pharyngeal motility and leading to delayed swallow initiation, diminished pharyngeal transport, and ineffective laryngeal protection.⁴⁷ Changes to the salivary glands may also become permanent, leading to xerostomia, which can contribute to reduced bolus formation and impaired timing of oral and pharyngeal transport.

Postoperative dysphagia is prevalent across a variety of procedures. Dysphagia occurs in approximately one-half of individuals following anterior cervical fusion,⁹⁹ and patients with multiple

surgical levels have a higher risk of dysphagia compared with those who only undergo fusion at one level.¹⁶ Decreased laryngeal sensation, pooling of secretions in the pharynx and larynx, impaired swallow initiation, and aspiration are common following endotracheal intubation and ventilator dependency. An inflated cuff does not eliminate aspiration.³² A tracheostomy tube can alter normal pharyngeal aerodynamics, eliminating the positive subglottic pressure normally associated with swallowing, which then hampers laryngeal protective reflexes. Tracheostomy occlusion through tracheostomy speaking valves, caps, or healing grants voluntary pharyngoesophageal clearance by allowing the buildup of subglottic pressure, which is necessary for an effective cough to help clear any penetrated materials.⁵⁶

It is also essential to be aware of the impact of medication regimens on swallow function. Many medications, including those available over the counter, can impact swallowing.⁸⁹ Medications may impair consciousness, reduce muscular coordination and sensory functions, or impact saliva production. Other medications bring on conditions such as mouth ulcerations, glossitis, esophagitis, and stomatitis. Some classes of drugs that are particularly concerning for negative impact on swallow function include nonsteroidal antiinflammatories, antibiotics, antihistamines, antispasmodics, anticholinergics, antimuscarinics, antihypertensives, anxiolytics, antipsychotics, anti-Parkinson disease agents, anticonvulsants, and even vitamins. In addition to medications that affect motor and sensory functions or damage mucosa, any medication that causes sedation can increase the risk of aspiration and impair airway clearance (Table 3.1).

Evaluation of Dysphagia

To effectively address swallowing disorders and reduce the risk for dysphagia-associated medical complications, the health care team must first complete a careful swallowing assessment using appropriate diagnostic procedures. The purpose of the swallowing evaluation is to assess dysphagia and, when appropriate, make recommendations for diet, swallow safety strategies, and rehabilitation interventions. A complete swallowing workup typically involves a bedside swallow screening followed by a clinical bedside

TABLE 3.1 Common Medications Affecting Swallow Function

Product Category	Examples	Possible Effects
Antibiotics	Penicillin, erythromycin, tetracycline	Glossitis, stomatitis, esophagitis
Anticholinergics	Benzotropine, ipratropium bromide, tiotropium bromide	Dry mouth, decreased appetite
Antidepressants	Amitriptyline, mirtazapine, escitalopram, sertraline, citalopram, fluoxetine	Drying of mucosa, drowsiness, possible weight gain
Antihistamines	Cough and cold medications	Drying of mucosa, sedative effects
Antipsychotics	Haloperidol, chlorpromazine, clozapine, risperidone	Tardive dyskinesia (movement disorder affects the face and mouth with repetitive, involuntary movements)
Diuretics	Furosemide, hydrochlorothiazide, spironolactone	Signs of chronic dehydration (dry mouth, thirst, weakness, drowsiness)
Mucosal anesthetics	Benzocaine	Suppresses gag and cough reflex
Sedatives	Phenobarbital, pentobarbital	Central nervous system depressant (drowsiness causing decompensation of patients with cognitive deficits and increases aspiration risk and impairs airway clearance)

evaluation and subsequent instrumental evaluation when indicated based on signs and symptoms of swallowing impairment identified during the clinical exam. The instrumental exam aims to determine the presence or absence of airway invasion objectively, clarify the relationship between symptoms and pathophysiology, and guide decision making regarding therapeutic intervention.

The two most common instrumental swallowing assessments across medical settings are the videofluoroscopic swallow study (VFSS) and the fiberoptic endoscopic evaluation of the swallow (FEES). In recent years, additional instrumental assessment tools, such as high-resolution pharyngeal manometry (HRPM) and computed tomography swallowing studies (CTSS), have emerged, giving clinicians more options for assessing swallowing function. Regardless of the type of assessment used to evaluate swallow function, the examiner should recognize that the results represent only a moment in time; therefore they need to be interpreted in the context of the patient's clinical presentation. Many variables, such as the underlying medical condition, medication regimen, cognitive status, age, feeding status, functional status, as well as unknown factors, can potentially influence function positively or negatively.⁵⁹ Moreover, it is not uncommon for individuals with dysphagia to demonstrate variable swallowing performance depending on the time of day because of fatigue or the timing of medications. The variability of swallowing performance throughout the day may be especially prominent for aging individuals and those with progressive neurologic diseases. Therefore it is imperative to interpret the results of the swallowing assessment in conjunction with the overall clinical picture.

Bedside/Clinical Swallow Assessments

Swallow Screenings

The swallow screening is usually the first step in identifying an increased risk of dysphagia. Swallow screenings may take on a variety of forms with a range of protocols, all with the purpose to accurately identify individuals who present a risk of dysphagia and expedite referral to a speech pathologist for further evaluation.¹⁵ Generally, the screening should be quick and minimally invasive, with the ability to be used by any team member. The swallow screening may identify risk factors for dysphagia using self-report, clinical history, and/or direct clinical observation.

Most swallow screening protocols include a brief history, examination, and swallowing challenge, typically swallowing water. The 3-ounce water swallowing test is common. Individuals are asked to consume 3 ounces of water uninterrupted without any overt signs of aspiration (i.e., cough, choking, throat clearing, wet voice quality). If the person cannot drink the entire amount uninterrupted or demonstrates signs of aspiration immediately after consuming the water, this is considered a "fail" on the screening, and the individual is considered at risk for aspiration. One commonly used tool that includes a brief cognitive screen and an oral examination in addition to the swallowing challenge is the Yale Swallow Protocol.⁵⁵

Other swallow screenings reported in the literature also use water with volumes ranging from 5 to 90 mL and different criteria for pass/fail.^{17,68,105,106} The implementation of swallow screening tools has resulted in a reduction of pneumonia rates,^{20,34} and screening tools that include swallow challenges yield the best patient outcomes.^{34,101} Furthermore, using water as part of a screening makes the screening relatively easy to administer.

Clinical Swallowing Evaluation

The clinical swallowing evaluation (CSE) is often conducted when signs and symptoms of dysphagia are identified by screening, patient report, or observation by a direct caregiver or family member. Although the CSE does not allow for any direct observation of the physiology of the pharyngeal phase of the swallow, other pertinent information can be obtained related to swallow function. The CSE has five components: (1) medical history review and patient/family interview, (2) cognitive/mental status function, (3) oral motor and sensory function, (4) laryngeal and pulmonary function, and (5) administration of oral swallowing trials.⁵⁹ Review of the patient's medical history and current medical status includes the medical course of treatment, admitting diagnosis, lab values, nutrition status, medications, and respiratory status. The clinician considers the impact of any surgical procedures and the results of any prior dysphagia assessments or treatments.

The cognitive-linguistic evaluation elicits information regarding the patient's level of alertness, orientation, memory (i.e., long-term, immediate, delayed), attention, reasoning, problem solving, and the individual's basic speech and language function. In addition to communication of basic needs, these skills also impact the patient's ability to follow directions and implement strategies for safe swallowing. For example, significantly increased impulsivity and distractibility as seen in right middle cerebral artery strokes can impact safety and subsequently raise the risk of aspiration considerably. Therefore all these elements must be factored into the speech-language pathologist's (SLP's) decision making and treatment planning as they relate to patient safety.

The primary focus of the oral exam is the evaluation of the jaw, lips, tongue, dentition, and hard/soft palate structures. These structures are assessed for any abnormalities with strength, symmetry at rest and upon movement, tone, atypical movement (e.g., apraxia), and structural deviations related to surgical/acquired condition or congenital factors. Preliminary studies have emerged demonstrating potential objective measures of oral strength and range that may be identified during the initial evaluation as predictors of swallowing outcomes.^{51,55} Dentition and condition of the oral mucosa are also evaluated during the oral motor assessment. If xerostomia is noted on the oral exam, rigorous oral care is indicated because colonization of the oral cavity with bacteria is exacerbated in people with poor salivary flow.

During the oral sensory component of the assessment, the clinician evaluates the individual's response to touch and temperature bilaterally for the chin, lips, tongue, inner cheeks, and palate. Attempts to elicit a gag reflex may reveal a sudden tonic reaction to a noxious stimulus in the posterior oropharynx. The absence of a gag reflex is not a clinical predictor of dysphagia, nor does it protect against aspiration because a subset of the normal population does not present with a gag reflex.⁵² In addition, pulmonary status is assessed through observation of the individual's respiratory rate at rest along with an assessment of volitional and reflexive (if elicited) throat clearing and cough production. Laryngeal function is indirectly assessed by evaluating the strength of the cough and vocal quality, vocal loudness, and pitch range. Nasal resonance is also assessed to indirectly determine the integrity of the velopharyngeal port.

The results of these initial components of the CSE will guide the clinician's decision to attempt or refrain from administering food and liquid trials. When attempting swallow trials, clinicians often start by asking the individual to produce a saliva swallow,

followed by presentation of single ice chips, and other bolus consistencies and volumes as deemed appropriate. The examiner will observe for oral phase deficits such as anterior spillage, buccal pocketing, prolonged mastication, and/or bolus propulsion through the oral cavity. The clinician may also palpate for laryngeal elevation by gently placing the fingers on the patient's neck at the level of the hyoid bone and thyroid cartilage to subjectively judge hyolaryngeal elevation during the pharyngeal swallow phase.⁵⁹ Furthermore, the clinician observes for any overt signs and symptoms of pharyngeal phase dysfunction, such as the presence of a cough, throat clear, or wet vocal quality either before, during, or immediately after the swallow. If the individual demonstrates any clinical signs of pharyngeal impairment during the swallow trials, a referral for an instrumental assessment of swallowing is indicated to objectively evaluate swallowing pathophysiology.

Instrumental Swallow Assessments

Videofluoroscopic Swallow Study

VFSS is a standard examination for the diagnosis and treatment of dysphagia, also known as modified barium swallow. It allows for a comprehensive oral, pharyngeal, and esophageal function evaluation. The procedure involves swallowing a contrast agent or food containing a contrast agent, while using real-time x-ray fluoroscopy to observe the movement of the swallowing structures and the flow of the contrast agent. This examination allows for the identification of both structural and functional impairments and assists in the determination of safe and appropriate interventions, such as whether oral intake is feasible or if alternative means of nutrition are necessary. It also involves decision making regarding compensatory strategies such as positioning, dietary modifications, and alternative swallowing methods for rehabilitation interventions.

Specifically, VFSS provides insights into oral motor function, bolus formation and propulsion, initiation of pharyngeal swallow timing, velopharyngeal closure, laryngeal closure, and UES opening. During this assessment, the clinician observes oral control and assesses pharyngeal clearance as well as airway protection, including laryngeal penetration and aspiration occurrences before, during, and after the swallow. VFSS is essential for detecting silent aspiration missed during CSEs and for guiding treatment decisions. The procedure typically occurs in radiology departments, involving collaboration between radiologists or physiatrists and SLPs. Clinicians trained in VFSS administration and interpretation ensure radiation safety during the procedure. VFSS protocols vary based on patient presentation, involving assessments with different bolus consistencies (i.e., thin, slightly thick, mildly thick, moderately thick, extremely thick liquids; regular, soft and bitesize, minced and moist, pureed, liquidized solids), sizes (e.g., 1, 5, 10, or 20 mL, and uncontrolled size), and presentation methods (e.g., syringe, cup, or spoon; self-intake or assisted), often introducing compensatory strategies to optimize swallowing safety and efficiency. The procedure aims to minimize radiation exposure while providing comprehensive insights into swallowing function. In the case of high-risk patients, careful consideration should be given to determine the maximum competence of the individual's swallowing function. Challenging the system and knowing the patient's maximum capability are helpful in understanding the underlying pathophysiology of the swallowing impairment and appropriate treatment directions.

The VFSS can also evaluate esophageal function. Although standardized evaluation methods have not yet been established, it is recommended that evaluation of esophageal function be performed in conjunction with oropharyngeal swallow function as much as possible. The contrast medium is ingested in the frontal view, and organic and functional abnormalities are observed from the flow of the contrast medium. The presence or absence of residuals, the location of residuals, and the state of esophageal peristalsis are observed. In rare cases, a tracheoesophageal fistula may be detected, so alternate fields of view may also be useful (Fig. 3.3).

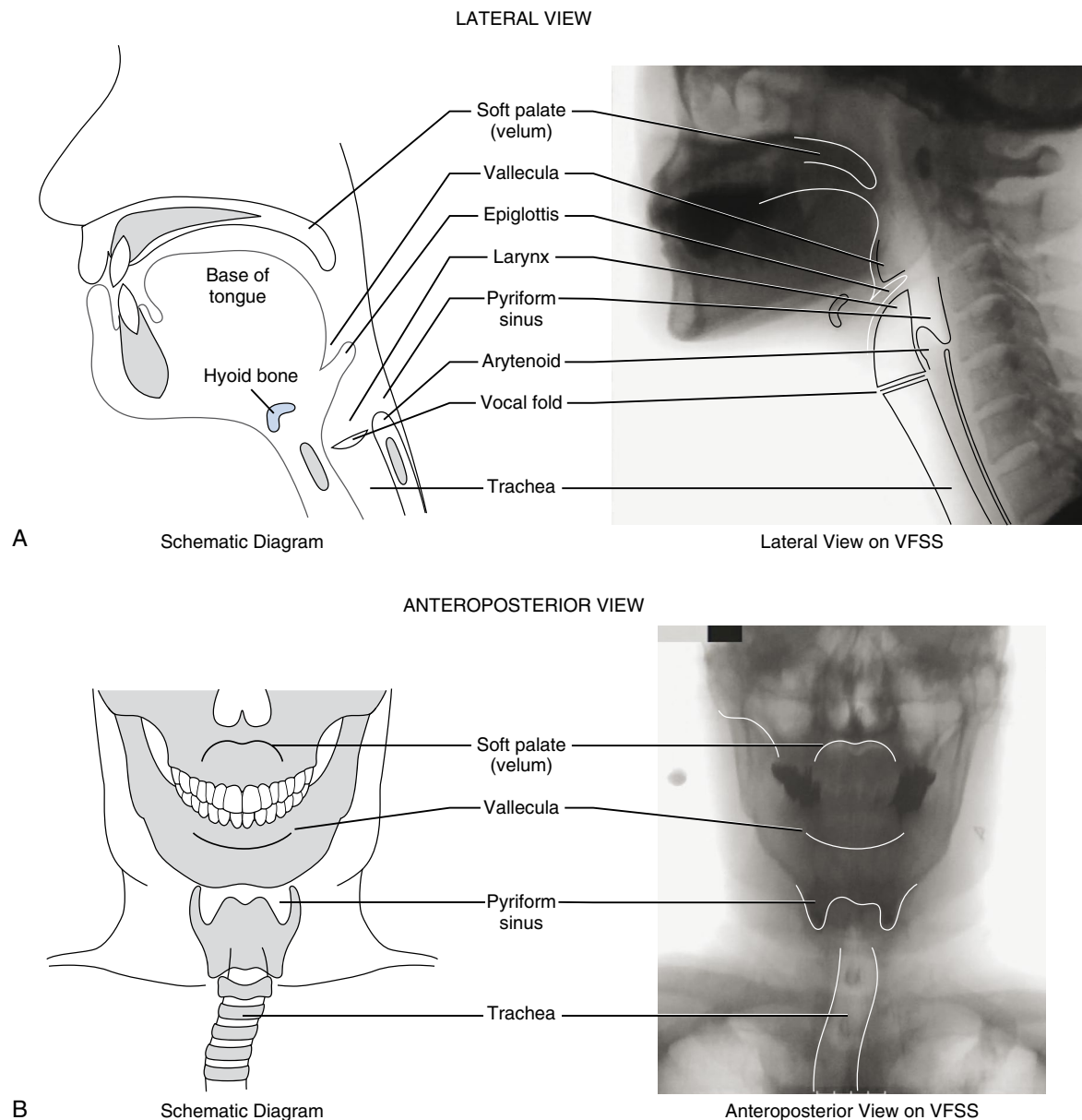
Fiberoptic Endoscopic Evaluation of Swallowing

FEES is a common instrumental assessment that allows for direct visualization of the pharynx and larynx, accumulated oropharyngeal secretion levels, swallowing function, and sensory ability in persons of all ages from infants to older adults. The flexible fiberoptic laryngoscope is equipped with a chip camera attached to a light source and video recorder, and the examiner (i.e., endoscopist) visualizes the image either directly through an eyepiece or on a monitor when using the camera attached to the laryngoscope. Images can be recorded for further analysis and playback for therapeutic/educational purposes.

To perform the procedure, the laryngoscope is passed through the nares following application of a topical anesthetic. Because of the placement of the laryngoscope, the oral phase and the esophageal phase of the swallow cannot be directly visualized during the FEES assessment. Additionally, there is a white-out period at the height of the swallow during which the pharyngeal squeeze at the level of the camera on the distal end of the scope causes light to reflect into the endoscope and obscure the view of the bolus and pharyngeal structures. For this reason, the examiner must make inferences based on observed pre- and postswallow components of the FEES.⁴⁹

The typical FEES protocol includes an assessment of the anatomic structures of the larynx at rest and in movement, the accumulated oropharyngeal secretion level, and bolus flow of various foods and liquids while swallowing. The examiner will instruct the individual to produce various speech and vocal tasks as well as a dry swallow before presenting foods and liquids that may be treated with food dye to enhance visualization of bolus flow during swallowing. If difficulty swallowing is observed, compensatory strategies (e.g., postural changes, swallowing maneuvers, and bolus modifications) may be introduced to identify optimal conditions for swallowing performance. Additional assessment of laryngopharyngeal sensation and airway protection is possible when performing the more extensive flexible endoscopic evaluation of swallowing with sensory testing. Sensory discrimination thresholds are determined by presenting pressure and duration-controlled air pulses from a port on the laryngoscope to the mucosa to elicit the laryngeal adductor reflex, thereby allowing objective sensory discrimination testing as well as testing for individuals with cognitive deficits who are unable to reliably respond to psychophysical testing.³

In addition to the ability to identify pooled oropharyngeal secretions, the amount of postswallow pharyngeal residue, and laryngeal penetration and/or aspiration, FEES is beneficial in other ways. For example, reduced vocal fold mobility has been associated with increased risk of aspiration,⁵³ and FEES allows for direct observation of vocal fold anatomy and movement. FEES may also be used as biofeedback in appropriate individuals

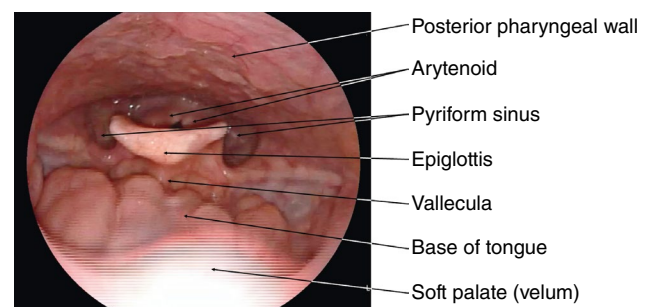


• **Fig. 3.3** Structures visualized using different views on a videofluoroscopic swallow study (VFSS). A VFSS permits visualization of swallowed contrast flow in relation to structural movement throughout the upper aerodigestive tract in real time; however, different views are optimal for the assessment of different structures and characteristics of the swallow. (A) Lateral view. (B) Anteroposterior view.

to confirm accurate performance and/or effectiveness of compensatory strategies (e.g., breath hold during a supraglottic swallow maneuver, pharyngeal clearance with effortful swallow). Although FEES is more invasive than VFSS, it can be done at bedside or in clinical treatment areas, and there is no radiation exposure (Fig. 3.4).

Computed Tomography Swallowing Study

Swallowing CT has significantly enhanced our ability to observe and assess swallowing dynamics, particularly regarding true vocal cords and UES opening, which were previously challenging to observe adequately in axial cross sections. Although the VFSS allows for visualization of swallow function in two dimensions,



• **Fig. 3.4** Structures visualized using fiberoptic endoscopic evaluation of the swallow. A fiberoptic endoscope is passed through the nose to visualize the larynx and pharynx as well as the movement of a food/liquid bolus in relation to structural movements.

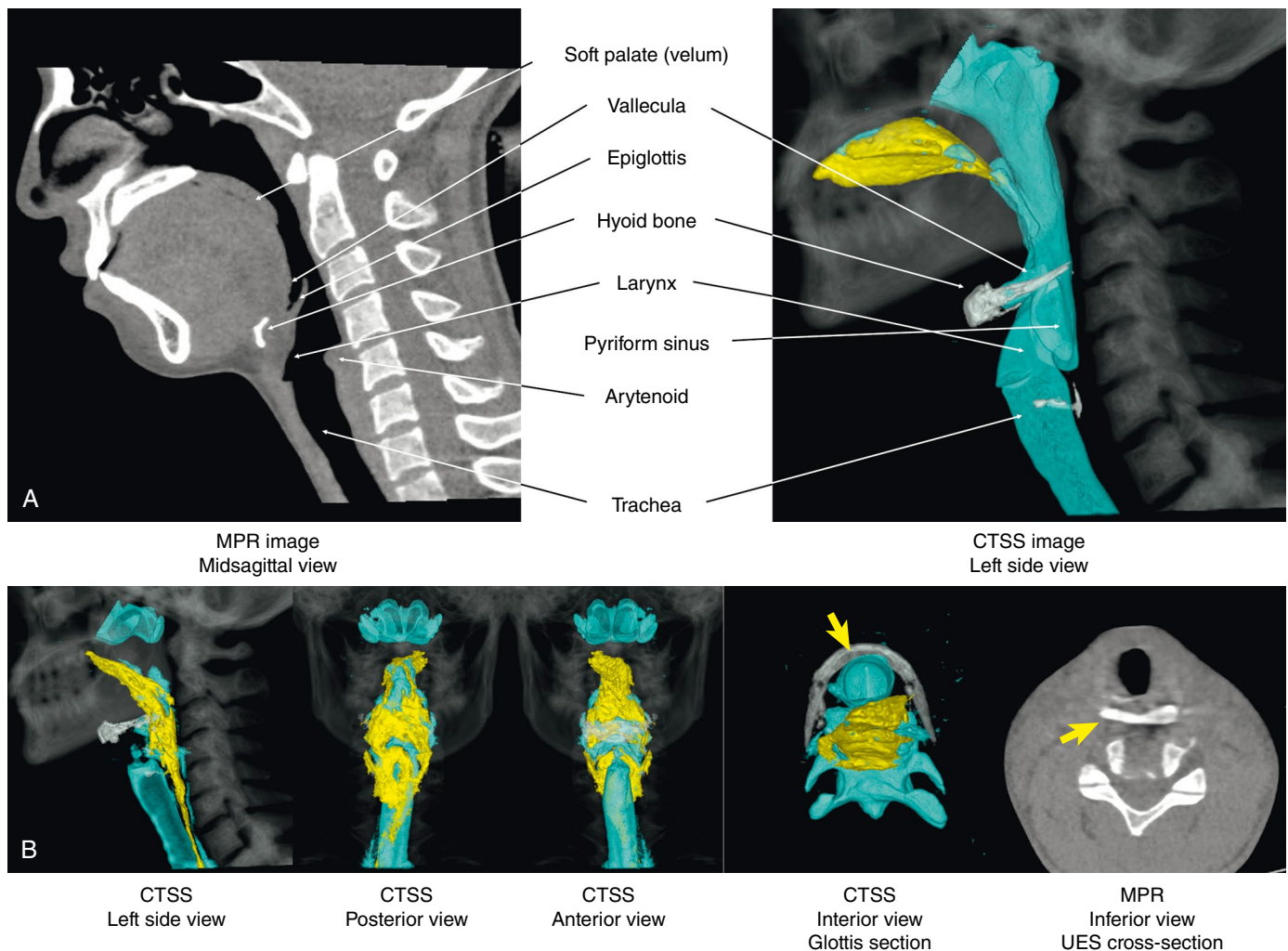
the CTSS has exceptional spatial analysis, allowing three-dimensional visualization of swallowing dynamics. Moreover, this three-dimensional view facilitates precise quantification of swallowing dynamics, enabling evaluation of pharyngeal constriction by volume and UES opening by cross-sectional area, thus improving accuracy in analysis. Swallowing characteristics that can be analyzed by CTSS include the onset, termination, and duration of movements of each swallowing structure, the volume of the pharyngeal cavity, the trajectory and distance of movements of the hyoid bone and larynx, as well as the area of the opening of the UES, and muscle length. However, despite its spatial prowess, temporal analysis remains limited to 10 frames per second, necessitating further advancement for tracking swallowing dynamics, which typically occur within 1 second. Additionally, while a 5-minute VFSS exposes individuals to 1.05 mSv of radiation, a 3.15-second CT exposes them to a relatively higher dose of 1.65 mSv, restricting the frequency and duration of scans. Thus a comprehensive approach that combines VFSS for an overview of swallowing pathology with CT for detailed evaluation is advisable^{37–39,44,82} (Fig. 3.5).

See [Videos 3.1](#) and [3.2](#) for examples of the modalities described above.

High-Resolution Pharyngeal Manometry

HRPM is a diagnostic test that provides a quantitative assessment of intraluminal swallowing-related pressure events. Introduced just over a decade ago¹⁰² as a means to evaluate swallow function, it involves the placement of a catheter with at least 36 solid-state circumferential pressure sensors, which are spaced at approximately 1-cm intervals to measure pressure events and bolus flow during the pharyngeal and esophageal phases of the swallow.⁷⁸ The pressure forces measured by a HRPM catheter are influenced by several factors, including the strength and sequencing of neural activation of swallowing musculature, luminal diameter, pharyngeal wall thickness, and the presence and compressibility of luminal content.

Individuals with dysphagia often present with weak but appropriately sequenced pharyngeal pressures, which indicates reduced force generation capacity of the pharyngeal musculature. In such individuals, HRPM can objectively diagnose pharyngeal



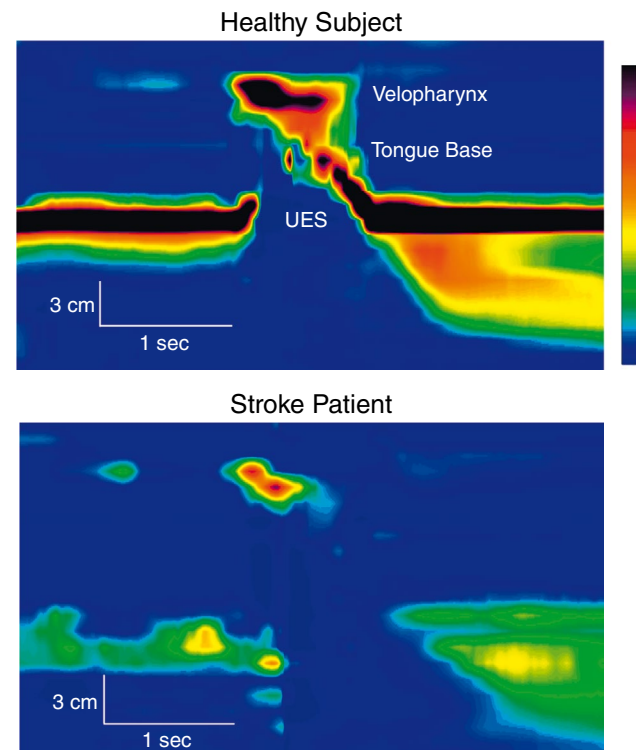
• **Fig. 3.5** (A and B) Structures visualized using computed tomography swallowing studies (CTSS) across multiple viewing planes. In these images, air column surface is indicated in *blue*, the swallowed bolus in *yellow*, and bone in *white*. MPR, Multiplanar reformation; UES, upper esophageal sphincter.

weakness and precisely identify the site of weakness by separating the contractile sequence into its component parts and potentially allow for more targeted therapeutic strategies aimed to augment pharyngeal force generation.^{42,43} In addition, impedance, or the measurement of electrical resistance, can be utilized in association to assist in the detection of postswallow residue¹⁷ and the characterization of potentially aberrant swallowing patterns such as piecemeal swallowing.⁸³

HRPM can also be used as a therapeutic biofeedback tool. With the HRPM catheter in place, individuals with dysphagia are given different types of food and liquid to swallow, during which swallow pressures are recorded and displayed visually for both the patient and clinician to view in real time. This information is used to create treatment protocols specific to each individual with targeted exercises to strengthen specific areas. With HRPM as biofeedback, adjustments can be made in swallowing and exercise execution in real time to improve effectiveness and improve function in low-pressure areas (Fig. 3.6; Table 3.2).

Treatment of Dysphagia

The goal of dysphagia treatment is to restore swallowing skills while continuously maintaining adequate nutrition, hydration, and safe (aspiration-free) swallowing for the individual. When planning treatment, the clinician must consider the individual's disease process and potential for recovery of function along with other important factors, such as motivation, ability to participate in a rehabilitation program, response to interventions, and availability of caregiver support. Early treatment of dysphagia has been shown to reduce the risk for aspiration pneumonia, reduce medical complications related to malnutrition and dehydration, and reduce the length of the hospital stay.⁴⁵ Equally important is the amelioration of barriers that decrease an individual's ability to participate in and enjoy the pleasures of eating orally. Small physiologic improvements in swallowing that do not result in an increase in quality of life miss the true goal of rehabilitation.¹²



• **Fig. 3.6** High-resolution pharyngeal manometry pressure map. Pressure is indicated by variation in color: *Blue* indicates low pressure, and *red* (almost black) indicates high pressure. (A) In healthy subjects, pharyngeal contraction causes the high-pressure zone to propagate from the velopharynx to the tongue base. At about the same time as pharyngeal contraction, the upper esophageal sphincter (UES) relaxes and drops from the high-pressure zone to atmospheric pressure, then reverses to the high-pressure zone with the end of swallowing. (B) Following stroke, velopharyngeal contraction is observed, but there is no high-pressure zone at the tongue base. UES relaxation is observed, but the UES pressure before and after swallowing appears very low compared with the healthy subject.

TABLE 3.2 Comparison of Diagnostic Modalities

	VFSS	FEES	CTSS
<u>Imaging Quality</u>	<u>Indirect Two Dimensional</u>	<u>Direct Two Dimensional</u>	<u>Indirect Three Dimensional</u>
Observable phase			
Oral	+	-	+
Pharyngeal	+	+	+
Esophageal	+	-	+
Radiation exposure	+	-	++
<u>Measurement Quality</u>	<u>Quantitative But With Limits</u>	<u>Qualitative</u>	<u>Quantitative</u>
Advantages	Two-dimensional visualization, available in most hospitals	Portable, clear visualization of vocal cord and soft tissues, can assess swallowing over an extended period of time, does not require contrast	Visualization in all planes, allows for assessment of volume (air and bolus)
Disadvantages	Not accessible in all settings, requires contrast	White-out period during swallow, transnasal passing of scope	Limited number of swallows because of radiation, requires contrast, available only for research in the United States

CTSS, Computed tomography swallowing studies; FEES, fiberoptic endoscopic evaluation of the swallow; VFSS, videofluoroscopic swallow study.

Dysphagia treatment approaches to improve the safety and efficiency of the swallow can be described in terms of compensatory or restorative strategies. The goal of compensatory swallowing strategies is not to change swallowing physiology but to prevent symptoms of dysphagia to maintain safety and ensure adequate nutrition and hydration. These techniques do not result in long-term physiologic changes and include strategies such as volume and texture modifications of foods and liquids as well as postural changes and maneuvers. Alternatively, restorative treatment involves swallowing exercises used to treat dysphagia by altering swallowing physiology and promoting long-term changes. Restorative exercises are expected to impact swallowing mechanics and bolus flow. Depending on the physiologic impairment identified by the instrumental swallowing exam, a combination of compensatory and restorative rehabilitation strategies may be implemented to both manage dysphagia symptoms and improve swallowing physiology.¹⁰⁷

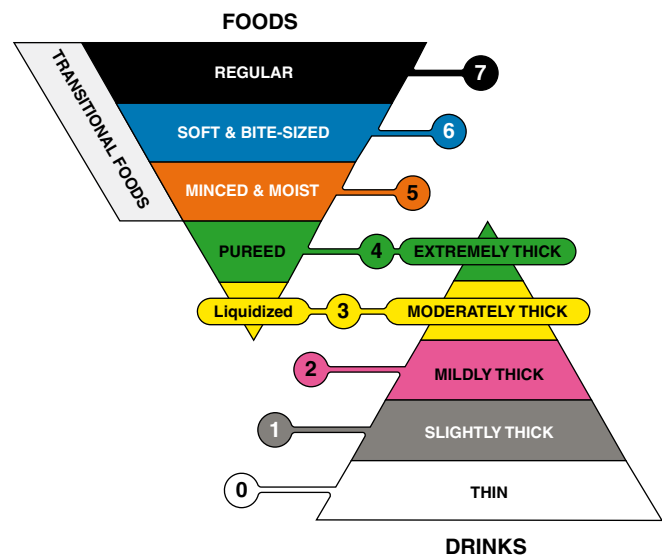
Compensatory Strategies in Swallowing Rehabilitation

Compensatory swallowing strategies are recommended when the objective of treatment is to prevent symptoms of dysphagia in order to maintain safety and ensure adequate nutrition, not change swallowing physiology. With a focus on improving quality of life, the goal is for the individual to regain the ability to participate in social activities and to enjoy the pleasures of eating and drinking through volume and texture modifications, sensory enhancement, and implementation of postural swallowing strategies.

Following a thorough assessment of swallow function, which typically includes an instrumental exam, the clinician must first determine if it is safe to feed the individual by mouth and in what capacity (i.e., begin with therapeutic oral trials vs. small or full meals). Decisions about recommendations for both diet modifications and swallowing maneuvers should be based on identification of impaired physiology observed during the instrumental exam.

Postural techniques alter the direction of bolus flow using gravity or changes in anatomic positioning during a swallow. These strategies should be tested and proven effective during instrumental evaluation before recommending their use. Examples include collapsing the pharyngeal space by turning the head toward the impaired side as well as performing a chin tuck to widen the valleculae and positioning the epiglottis in a more protective posterior position to reduce laryngeal penetration or aspiration.

Sensory enhancement techniques are also part of the treatment strategies. These interventions aim to increase the sensory input to the brainstem, thereby maximizing the swallow response. For example, taste and chemesthesis can increase oral and pharyngeal transit in individuals with dysphagia.⁸⁸ A sour bolus has been found to elicit earlier onset of lingual activity to propel a bolus to the oropharynx and trigger a pharyngeal motor response in individuals with stroke or neurologic disorder.⁶¹ The application of thermal tactile stimulation or thermal gustatory stimulation is somewhat controversial, but some data suggest that it may increase corticobulbar activity, which has been associated with recovery of swallow function following stroke.⁹⁰ The clinician uses a chilled laryngeal mirror or iced lemon glycerin swab to stroke the anterior faucial arches in the posterior oral cavity to increase sensory input, thereby facilitating a timely swallow response. These strategies have been shown to result in immediate changes in swallowing function, but long-lasting effects have not been proven (Fig. 3.7).



• **Fig. 3.7** The International Dysphagia Diet Standardization Initiative (IDDSI) framework. The IDDSI framework consists of a continuum of eight levels where drinks are measured from levels 0 to 4 and food is measured from levels 3 to 7. In this way, the framework provides a common terminology that can be used across users and settings to describe drink viscosity and food textures. (Courtesy The International Dysphagia Diet Standardisation Initiative 2019 at <https://iddsi.org/framework>. Licensed under the Creative Commons Attribution Sharealike 4.0 License.)

Modification of food and liquid properties (texture and viscosity) as well as presentation characteristics (volume and temperature) are common therapeutic approaches, although most clinicians agree that this should be the treatment of last resort. It is important to standardize terminology and definitions to describe texture-modified foods and liquids and to maximize the reliability of diet modifications across settings, practitioners, and cultures. To this end, the International Dysphagia Diet Standardization Initiative (IDDSI)¹¹ was formed with the aim to improve the lives of people with dysphagia by standardizing the classification and description of food and liquid textures. The IDDSI framework consists of a continuum of eight levels (0 through 7), where drinks are measured from levels 0 through 4, and foods are measured from levels 3 through 7. Liquids range from thin (level 0) to extremely thick (level 4), while foods range from regular/easy to chew (level 7) to liquidized (level 3). In addition, the IDDSI framework offers a standard approach to caregiver training by means of a flow test to ensure that thickened liquids are mixed to the appropriate viscosity, eliminating the confusion posed by the variability of methods previous utilized to thicken liquids.¹¹ IDDSI diet levels have been adopted by many medical settings across the world.

For individuals with decreased oral motor strength and/or coordination resulting in difficulty with bolus formation and/or bolus transit through the oral cavity, a diet texture change to soft/bitesize, minced and moist, or pureed may increase the efficiency and/or safety of swallowing. Many individuals with neurogenic dysphagia are at greatest risk of aspiration with thin liquids because of poor bolus control, reduced lingual propulsion, delayed pharyngeal swallow, or a combination of these impairments. In cases such as this, thickened liquids (slightly thick, mildly thick, moderately thick, or extremely thick) may be recommended to minimize the risk of aspiration.

Restorative Strategies: Exercise Training and Plasticity Considerations

The oropharyngeal system exhibits considerable plasticity in the adaptation of new tasks or sensory experiences in health and disease. Therefore the goal of rehabilitative exercises for individuals with dysphagia is to change and improve the physiology of swallowing in force, speed, and/or timing to promote long-term treatment effects. In this way, retraining of the neuromuscular system results in neuroplasticity.¹⁰ For example, the proportion of neurons in the primary somatosensory cortex (S1) and primary motor cortex (M1) that are correlated with tongue protrusion increases following daily tongue training. Additional studies have shown the importance of task-related specificity among tongue motor representations and the cortical masticatory area.⁹³ The specificity in orofacial gestures and swallowing for targeted neuroplasticity interventions suggests that consideration be given for stimulus salience (nonswallow task training compared with swallow task training) to affect beneficial behavioral changes and adaptive plasticity for swallowing and oral motor behaviors.⁷⁷

A variety of exercises exist ranging from direct to indirect and those that incorporate swallowing or nonswallowing exercises to effect change. Many of the swallowing exercises follow the principles of neuromuscular rehabilitation, including use it or lose it, training specificity, repetition, muscle overload, and training intensity to facilitate increased muscle strength.^{50,91} For example, the strength training principles of overloading and specificity have been successfully applied to swallow training. Following the principle of specificity, it is best to select exercises that are similar to the target behavior.³⁰ The Mendelson and tongue-hold (i.e., Masako) exercises apply this principle as the individual is attempting swallowing function during these exercises. Overloading involves completing exercises at resistance levels higher than typically used. During the Mendelson maneuver, the swallow is prolonged, lasting longer than the normal duration. When completing the tongue-hold maneuver, the anterior tongue is anchored, resulting in increased pharyngeal constrictor activity. Examples of exercise training that incorporates increased intensity of resistance are the isometric lingual exercise, expiratory muscle strength training (EMST; a technique targeting the respiratory musculature), effortful swallow, Mendelson maneuver, tongue-hold maneuver, and Shaker exercises. The importance of repetition of these exercises to induce change in swallow function must also be emphasized. Extensive and prolonged practice improves neural substrates and serves to assist in both the acquisition and maintenance of improved swallow function.

Although the body of evidence to support the use of specific exercise training has been growing tremendously in recent years, not all exercises have sufficient evidence acquired through well-designed large studies to support their use. Some interventions have proven effective in healthy normal individuals or may have been effective in a small number of study participants and therefore may warrant further investigation to assist in guiding clinicians to consistently use evidence-based practice. Additionally, individuals with cognitive dysfunction may require ongoing 1:1 or close supervision and cueing to perform the exercises and multistep maneuvers effectively. Use of visual biofeedback by endoscopy, EMG, FEES, or other means may serve to positively impact accurate performance of therapeutic exercises, such as the supraglottic swallow technique, and increase effectiveness of the exercise program.

Although nonswallowing exercises do not meet all the principles of neuroplasticity (i.e., specificity and overload), some have been shown to have good evidence for their efficacy. For example,

the Shaker head lift exercises use a combination of isometric and isokinetic exercises to improve the strength of the suprahyoid muscles over time and increase the opening of the UES. Consistent performance of this exercise has resulted in favorable long-term effects in individuals with dysphagia. EMST has been shown to improve both ventilatory and nonventilatory functions and has reported positive effects on swallowing.⁵⁰

Other nonswallowing exercises include the Lee Silverman Voice Treatment (LSVT), chin tuck against resistance (CTAR), and lingual strengthening exercises. LSVT is a proven program used to improve vocal intensity in persons with Parkinson disease; however, it has been found in pilot studies to produce gains in swallowing and cough function caused by a shared neurologic network.⁹⁵ CTAR is an exercise that targets the suprahyoid muscles to improve function of the muscles that comprise the UES. A pilot study compared the use of CTAR exercise in the treatment group of subacute stroke patients with a traditional therapy regimen received by the control group; it revealed that CTAR treatment resulted in significantly improved laryngeal elevation and epiglottic closure as well as pharyngeal clearance and greater airway protection in the experimental group.⁸⁷ Finally, lingual exercises have been investigated using the Iowa Oral Performance Instrument, a device that measures the maximal amount of tongue pressure a person can produce by lingual compression of a tongue bulb against the hard palate. Lingual exercises have been shown to have short-term effects in healthy normal participants but mixed results for swallowing safety and efficiency.⁹² Further study is warranted for all nonswallowing exercises to determine whether there is long-term benefit across dysphagia populations.

The use of neuromuscular electrical stimulation (NMES) to improve muscle function is not new in rehabilitative medicine but has grown in use as a dysphagia treatment method over the past 2 decades. It has been hypothesized that the use of surface electrodes taped to the skin overlying the submental or anterior cervical strap muscles (59 seconds on, 1 second off, for 60 continuous seconds) will result in strengthened swallowing musculature and a heightened sensory feedback loop for swallowing pathways.²⁹ However, contradictory findings have been reported with one study of individuals with chronic pharyngeal dysphagia showing that electrical stimulation actually depressed the hyolaryngeal complex in those with little to no hyolaryngeal movement, placing individuals at increased risk of airway invasion.⁶² Additional research with a focus on dosage, timing, and applications in specific patient populations is needed to determine whether NMES coupled with swallowing therapy may result in beneficial effects (Table 3.3).

Other Interventions for Dysphagia

Surgery is seldom indicated for the treatment of oropharyngeal dysphagia. In the case of UES or LES stenosis, dilation can be indicated. Surgery is also performed for Zenker diverticulum when the diverticulum is large enough to interfere significantly with eating and drinking.¹⁰⁹ Cricopharyngeal myotomy is a procedure that disrupts the cricopharyngeus muscle for the purpose of reducing UES pressure. This decreases the resistance to flow from the pharynx to the esophagus. However, evidence of its effectiveness is limited.¹⁴ Botulinum toxin injections are occasionally used to treat dysphagia, especially in cases of oromandibular or lingual dystonia, trismus, or cricopharyngeal dysfunction with failure of UES relaxation.¹¹⁰ In cases of severe dysphagia, permanent tracheostomies in combination with laryngectomy or another procedure to close the larynx can serve to separate the airway from the pathway of food

TABLE 3.3**Summary of Traditional Compensatory and Restorative Dysphagia Strategies**

Intervention	Therapeutic Approach		Physiologic Target		Physiologic Effects	Clinical Indications for Use
	Compensatory	Restorative	Oral	Pharyngeal		
Bolus modification	+		+	+	Increased viscosity: ↑ lingual-palatal contract pressure, pharyngeal pressure and upper esophageal sphincter (UES) relaxation, slows bolus transit. Increased bolus volume: ↑ bolus transit time, sustains laryngeal elevation and hyoid excursion; ↑ laryngeal closure duration and UES opening; ↓ duration of tongue base contract to posterior pharyngeal wall Other options: ↓ bolus volume, modify food texture	Impaired mastication; anterior spillage; laryngeal penetration; aspiration; pharyngeal residue
Breath hold (i.e., supraglottic swallow, super-supraglottic swallow)	+	+	+	+	↑ airway closure by increasing arytenoid approximation and true vocal cord closure; ↑UES opening during the swallow; early airway protection; prolonged hyolaryngeal excursion	Laryngeal penetration/aspiration during the swallow; delayed swallow initiation
Chin tuck	+		+		Expansion of vallecular recesses; approximation of tongue base toward pharyngeal wall; narrowing entrance to the laryngeal vestibule, ↓ in distance between hyoid and larynx; ↑ duration of swallowing pause	Premature spillage; pharyngeal residue in valleculae; laryngeal penetration/aspiration attributable to delayed swallow response
Effortful swallow	+	+	+	+	↑ hyolaryngeal excursion, duration of hyoid elevation, and UES opening, laryngeal closure, lingual pressures, peristaltic amplitudes in the distal esophagus, and pressure and duration of tongue base retraction	Pharyngeal residue in valleculae or pyriform sinuses; ↓ airway closure
Head turn (one side only)	+			+	(Toward side of impairment) Redirects bolus to side opposite rotation; ↓ UES pressure on side opposite rotation; narrows laryngeal entrance and ↑ vocal fold closure	Unilateral laryngeal/pharyngeal residue; reduced UES opening; unilateral vocal fold impairment
Head rotation (to either side)	+			+	↓ UES pressure on the side opposite to the head turn allowing for ↑ extension and duration of UES opening	Pharyngeal residue at the level of the UES; reduced UES opening
Head tilt	+		+		Head tilt to the stronger side (ear to shoulder) diverts the bolus toward the stronger side of the oral cavity and pharynx	Unilateral pharyngeal residue; unilateral oral weakness
Liquid wash; alternating liquids and solids	+		+	+	Clears oral and pharyngeal bolus residue	Oral residue; pharyngeal residue
Masako (tongue hold)		+		+	↑ anterior bulging of the posterior pharyngeal wall	↓ tongue base and pharyngeal wall contract
Mendelson	+	+		+	↑ extent/duration of hyolaryngeal excursion, UES opening, pharyngeal peak contractions, bolus transit time/duration, and pressure of tongue base contract	↓ hyolaryngeal elevation/excursion and/or ↓ duration of UES opening resulting laryngeal penetration or aspiration; pharyngeal residue at UES

TABLE 3.3 Summary of Traditional Compensatory and Restorative Dysphagia Strategies—cont'd

Intervention	Therapeutic Approach	Physiologic Target	Physiologic Effects	Clinical Indications for Use	
Multiple swallows	+	+	+	Cued or spontaneous dry swallow clears residue after primary bolus swallow	Oral residue; pharyngeal residue
Oral hold	+			↓ premature spillage and swallow response delay	Laryngeal penetration or aspiration before the swallow caused by premature spillage or delayed swallow response
Shaker exercise		+	+	↑ anterior hyolaryngeal excursion, UES opening; strengthens suprahyoid muscles; enhances thyrohyoid shortening	↓ UES opening; weakness of suprahyoid muscles
Throat clear	+		+	Expels airway invasion	Laryngeal penetration; aspiration

TABLE 3.4 Summary of Alternative Means of Alimentation

Feeding Modality	Insertion	Indication	Possible Complications
Nasogastric tube	Inserted transnasally at bedside; confirmed via stethoscope or x-ray film	Short-term nutrition and hydration	Easily dislodged; can lead to ulceration and stricture
Orogastric tube	Inserted transorally at bedside; confirmed with stethoscope or x-ray film	Short-term nutrition and hydration; unable to use nasal passage	Easily dislodged; can lead to ulceration and stricture
Gastrostomy tube	Generally inserted endoscopically	Long-term nutrition and hydration	Infection, bleeding, perforation, clogging, and aspiration
Jejunostomy	Inserted surgically into small intestine	Absence of or inability to use the stomach; severe reflux aspiration	Clogging, diarrhea; questionable benefit for reducing aspiration
Parenteral nutrition	Intravenous (can be central peripheral)	Inability to use the digestive system; severe malnutrition; transition to a long-term nutrition method	Infection, gastrointestinal atrophy, blood clots, glucose imbalances, fluid overload

and liquids. Although this definitely prevents aspiration related to swallowing, this approach has the serious consequence of preventing phonation and can still result in significant dysphagia.¹⁰⁴

Bypassing the Oropharyngeal Swallow Process

When adequate and safe oral intake is not possible, pharyngeal bypass procedures eliminate the need for swallowing and provide alternative means for achieving nutrition, hydration, and medication administration.⁴⁶ In addition to dysphagia with NPO status, some individuals with dysphagia experience extended mealtimes because of a variety of factors, such as physical changes, changes in taste perception, anxiety related to oral intake, and cognitive impairment. Extended mealtimes (e.g., food intake periods lasting more than 30 minutes) often result in reduced caloric intake, which can negatively impact nutritional status. In these cases, supplemental means

of nutritional intake are indicated. Short-term feeding options include nasogastric and orogastric tubes, which are typically used when alternative nutrition will be required for 6 weeks or less. Gastrostomy and jejunostomy tubes are often used in cases of severe dysphagia when alternative feeding methods will be required long term. Although the use of feeding tubes is generally safe, they do not necessarily prevent aspiration, and tube feeding may in fact promote gastroesophageal reflux with secondary aspiration of stomach contents. In one study, pneumonia developed in up to 44% of patients with acute stroke who were fed with gastrostomy. Furthermore, 66% of those patients continued to aspirate chronically, with mortality rates greater than 50% after 1 year.¹⁸ Measures to minimize aspiration include head elevation, good hygiene with bag and line handling, use of slow and continuous feeding rather than bolus, and monitoring for gastric residue. Parenteral nutrition, or feeding intravenously, is often given temporarily to those who need an immediate

boost of calories before transitioning to a longer-term nutrition solution, those with malnutrition, those on noninvasive ventilation modalities that may increase risk of aspiration such as high flow nasal cannula, and/or those in which the digestive system is not functioning. Parenteral nutrition can be partial or total and is sometimes classified by the type of vein used to deliver the nutrition as in the case of central parenteral nutrition (delivered through a central vein) (Table 3.4).

Summary

In summary, swallowing is a complex sequence of voluntary and reflexive movements comprised of multiple phases mediated by cortical and subcortical regions of the brain. Impaired swallowing function can arise from a wide variety of functional and structural deficits resulting in sensory and/or motor impairment. Consequences of these impairments can lead to serious complications including dehydration, malnutrition, pneumonia, or airway obstruction which is why it is important to approach therapeutic intervention with a patient-centered, multi-disciplinary plan of care. This most often includes instrumental assessment to accurately identify biomechanical components of the individual's swallow function to target for compensatory or restorative interventions with the goal of maintaining safety while maximizing function and quality of life.

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